

Assignment 7

Trapezoidal rule

Numerical Integration Using the Trapezoidal Rule

Problem: Evaluate the integral $I = \int x^2 dx$ (from 0 to 2) using the Trapezoidal Rule for $n = 1, 2, 4, 8, 16$ trapezoids, and find the absolute error.

Exact Value: $I = [x^3/3]$ (evaluated from 0 to 2) $= 8/3 = 2.6666667$

Trapezoidal Rule: $I_t = (h/2) [f(x_0) + 2\sum f(x_i) + f(x_n)]$ where $h = (b - a)/n = 2/n$ and $f(x) = x^2$.

Absolute Error: $\text{Error} = |I - I_t|$

Results:

No. of Trapezoids (n)	I_t (Trapezoidal Approximation)	Absolute Error $ I - I_t $
1	4.000000	1.333333
2	3.000000	0.333333
4	2.750000	0.083333
8	2.687500	0.020833
16	2.671875	0.005208

Example for $n = 2$:

$$h = (2 - 0)/2 = 1$$

$$I_t = (1/2)[f(0) + 2f(1) + f(2)] = (1/2)[0 + 2(1) + 4] = 3$$

$$\text{Error} = |8/3 - 3| = 1/3 = 0.333333$$

Additional Trapezoidal Rule Problems

2. A Function Without an Elementary Antiderivative

Consider $I = \int e^{-x^2} dx$ (from 0 to 1). Use the composite trapezoidal rule with $n = 4, 8, 16, 32$.

Reference value for comparison: $I \approx 0.7468241328$.

For each n , $h = 1/n$ and the composite trapezoidal formula is used.

n	h	I_t	Absolute Error
4	0.250000	0.7429840978	0.0038400350
8	0.125000	0.7458656148	0.0009585180
16	0.062500	0.7465845968	0.0002395360

32	0.031250	0.7467642547	0.0000598782
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Observation: For these values of n, increasing n improves the approximation: the values approach the reference value and the absolute error decreases.

3. Data Instead of a Function

Given the experimental data:

x	0	0.5	1.0	1.5	2.0
f(x)	1.00	1.65	2.70	4.48	7.39

Here $h = 0.5$. Applying the composite trapezoidal rule:

$$I_t = (0.5/2)[1.00 + 2(1.65 + 2.70 + 4.48) + 7.39]$$

$$I_t = 6.5125$$

Why numerical integration is useful:

the function is available only as discrete experimental data points, so an analytical antiderivative may not be available.

The trapezoidal rule estimates the integral directly from the measured values.